

# Hydrochemistry evolution mechanisms of groundwater in Hancheng mining area over the past 52 years: Induced by hydrodynamic changes

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## Research Article

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# Abstract

Long-term over-exploitation of groundwater cause significantly water table decline and water quality degradation, understanding the hydrochemistry evolution mechanism of groundwater in mining areas is significant for the rational exploitation and protection of groundwater. A combination of statistical analysis and hydrogeochemical simulation was applied to study the temporal variation of hydrochemistry of Ordovician limestone water (OLW) in Hancheng mining area, and reveal the influence of hydrodynamic changes. Statistical analysis shows that the TDS of OLW in the of Hancheng mining area had increased about 212% during 1971 to 2022, of which most was contributed by  $\text{Na}^+$ ,  $\text{Cl}^-$ , and  $\text{SO}_4^{2-}$ . Combining with the dynamic change of the flow field, the increased mixing proportion of the deep buried OLW contributes to the increment of TDS. The mixing modeling of PHREEQC software were used to calculate the mixing ratio of the surface water and the deep OLW. The result showed that the OLW was mainly recharged by surface water in phase I, which accounts for 94% for the southern zone and 62% for the northern zone; while it declined to 37% and 55% in phase II, 27% and 36% in phase III. Accordingly, the recharge ratio of deep buried OLW is continuously increase, which accounts for 6% for the southern zone and 38% for the northern zone in phase I, and increased to 63% and 45% in phase II, 73% and 64% in phase III. The results of the mixing simulation verified that the increasing recharge proportion of deep groundwater is the key reason for the water quality deterioration in Hancheng mine area. Inverse modeling was conducted to recognize the main hydrogeochemical processes, the dissolution of dolomite, gypsum, halite, the precipitation of calcite and cation-exchange dominated the evolution of major-ion concentration in the OLW along SE-NW direction. This study will contribute to the future management of groundwater in mining area.

## 1 Introduction

Global groundwater abstraction has increased from  $\sim 312 \text{ km}^3$  in 1960 to  $734 \text{ km}^3$  in 2000 (Wada et al., 2010) and  $1400 \text{ km}^3$  in 2016 (Turner et al., 2019). Growing water demand and extreme climate events have intensified freshwater scarcity, exacerbating issues like groundwater depletion, freshwater salinization, and shortages in irrigation and ecological water supplies (Yang et al., 2024; Liang et al., 2021; Liang et al., 2024; Tian et al., 2025), especially in coal mining area where groundwater dewatering is intensive while the ecosystem is fragile (Ma et al., 2020; Guo et al. 2022; Li et al., 2021; Shi et al. 2018).

Karst aquifers constitute one of the world's largest freshwater reserves (Wu et al., 2025; Goldscheider et al., 2020), the estimated number of karst water consumers in 2016 was 678 million or 9.2% of the world's population (Stevanović, 2019). Karst water is also an indispensable resource for ecosystems, agriculture and economic activities (Tunde Olarinoye, 2020). Karst water makes great contributions to the growth of cities and development of industry in the water-scarce north China (Udimal et al., 2017). The once high-quality Karst groundwater now faces challenges of over-extraction and declining quality after long-term over-exploitation by industry and dewatering during mining (Tiwarly 2001; Younger 2004; Singh et al. 2011; Liang et al. 2013; Ali et al. 2017; Yin et al., 2016), which has been challenging the sustainable development of the region, especially under situations of climate change and extreme droughts in recent years (Qi et al., 2025; Zhou et al., 2025). Investigating the Karst water hydrogeochemical evolution mechanism of groundwater in

mining area can serve decision-makers and researchers to develop sustainable plans to preserve Karst water resource in the future.

There have been extensive research findings on the hydrochemical evolution of Ordovician limestone water. Most study thought that along the flow path, the water primarily undergoes precipitation (or dissolution) of calcite and dolomite, dissolution of gypsum and halite, as well as cation exchange (Qiao et al., 2019; Xue et al., 2023). However, this conclusion was drawn solely based on hydrochemistry basis, without clarifying the hydrodynamic changes, namely whether the flow path occurs under natural conditions or within a cone of depression due to over-exploration and intensive dewatering, which are the common situations in mining area (Chen et al. 2020; Sun et al. 2022, 2023 and 2024; Wang 2022). Therefore, further research on hydrochemical evolution should be conducted with consideration for the evolution of the hydrodynamic field. In addition, some researches combined hydrochemistry with hydrodynamics to reveal hydrogeochemical evolution process (Qu et al., 2023; Li et al. 2024; Zhang et al. 2024). While the evolution of hydrogeochemical characteristics in groundwater is a protracted process, long-term monitoring is required to reveal its developmental patterns. Consequently, investigating the long-term effect of hydrodynamic field change on the hydrogeochemical evolution becomes fundamental for understanding the hydrogeochemical evolution mechanism, which is the theoretical basis of water resources protection (Chen et al. 2021; Qiu et al. 2021; Yang et al. 2021).

The analysis of groundwater components, which are dominated by mineral dissolution (precipitation), decarbonation, desulphidation, ion exchange and mixing et al., is effective to characterize the hydrochemical processes (Wu et al. 2018; Zhang et al. 2020; Xiao et al. 2018; Yang et al. 2021). Hydrochemical methods, such as ion ratios, multivariate statistical analysis, isotopes tracing, hydrochemical modeling, etc., have been widely used to analyze the hydrochemical evolution (Guo et al. 2016; Zhang et al. 2016; Liu et al. 2019; Wang et al. 2019; Hao et al. 2020; Guo et al. 2020). Based on the analysis of groundwater components and hydrochemical methods, this research focus on the temporal variation of chemical compositions of OLW in mining area during long-term period to reveal the hydrogeochemical evolution mechanism under conditions of hydrodynamic change induced by over- exploitation and intensive dewatering.

The Hancheng Mining Area is one of the representative coal mining zones of the North China-type coalfield, characterized by complex hydrogeological conditions. The most significant threat comes from the Ordovician limestone aquifer, which underlaid the coal-bearing strata. Known for its substantial thickness and high water-yield capacity, this aquifer was once a crucial water supply source. However, decades of intensive mining operations have led to continuous decline in water levels and progressive deterioration of water quality. Previous research on the OLW has primarily focused on mine water inrush (Qiu et al. 2020; Qiao et al. 2017; Cao et al. 2022), while paying insufficient attention to the evolution of its hydrodynamic field under conditions of prolonged over-exploitation. However, understanding such hydrodynamic evolution is fundamental to revealing the evolution of its hydrochemical field. Therefore, this study specifically investigates the evolution of the hydrodynamic field under prolonged extraction and its induced hydrochemical evolution. Its goals were to: (1) identify and quantify the hydrodynamic field change of OLW using numerical simulation method; and, (2) understanding the temporal evolution of chemical compositions of OLW using statistical analysis; (3) analyze the mechanism of hydrogeochemical evolution based on hydrodynamic field change. A conceptualized flowchart with key components in this study is (Fig. 1):

## 2 Overview of the study area

### 2.1 Landform and hydrology

Hancheng mining area is located in the eastern margin of Hancheng city which lies on the west bank of the Yellow River. The Yellow River serves as the eastern boundary of the Weibei Plateau which is located in the middle of Shaanxi Province, China. The annual average temperature is 13.5°C, with maximum temperature of 42.6°C and minimum of -14.8°C, and an average annual precipitation of 559.7 mm. The regional topography features higher elevations (> 1000 m) in the northwest, characterized by mountainous terrain, while the southeast lies at lower elevations (< 400 m) within the Yellow River valley (Fig. 2). These distinct geomorphic units are separated by the Hancheng Normal Fault ( $F_1$ ) which was formed during the Himalayan period under NW-SE (northwest-southeast) tensional stress.

### 2.2 Geology and hydrogeology

Before the formation of the  $F_1$ , the study area was influenced by the Yanshanian SE-directed thrust nappe tectonics, causing steeply dipping, followed by folding and faulting of the strata that formed the reverse fault ( $F_2$ ) in the northwestern side of  $F_1$ . Consequently, the strata dip northwestward, resulting in the Ordovician aquifer deep buried in the northwestern area and outcropped in the southeastern side (Fig. 3), where the aquifer is dissected in different incision depths by four rivers: the Yellow River, Zaokai River, Pan River and Jushui River (Fig. 2). Tectonic movements and cyclic sea level fluctuations led to repeated deposition of marine and terrestrial sediments. In the lower part of the sequence biochemical and chemical sediments such as limestones, dolomites and gypsum predominate, while in the upper part, clastic sediments such as sandstones predominate.

Structurally confined, the OLW receives exogenous recharge through meteoric infiltration and fluvial leakage at its southeastern outcrop belt. (zone Ⅰ in Fig. 4), then flows northwestward along the strata inclination, exhibiting progressively stagnant flow behavior with increasing depth (zone Ⅱ in Fig. 4). Alternatively, after been recharged it may migrate along the strike (NE direction) of the fault zone of  $F_1$  and  $F_2$  before ultimately discharging into river valleys. While a transitional belt (zone Ⅲ in Fig. 4) between the steeply inclined (zone Ⅰ) and gently inclined zones (zone Ⅱ), where the flow flux and direction depend on the difference in hydraulic head which exist across the accommodation zone. When the hydraulic head in zone Ⅰ is higher (e.g. in rainy seasons) than that in section Ⅱ, water flows to the northwest, otherwise (e.g. in intensively exploitation period) the water flows to the southeast. According to the previous hydrogeological investigation, Hancheng mining area can be divided into two sub-hydrogeological units, the northern zone and the southern zone, bounded by the Pan River (Fig. 2).

### 2.3 Extraction history

The Hancheng mining area covers approximately 110 km<sup>2</sup>, hosting multiple coal mines, power plants, steel mills, and chemical plants, most of which are concentrated in the southeastern zone of the mining area. These facilities predominantly extract shallow-buried OLW as their primary water source, with exploitation

activities commencing in the late 1970s. Among these facilities, the Hancheng Power Plant pioneered large-scale extraction of OLW, maintaining an extraction rate of approximately 1,452 m<sup>3</sup>/h continuously for 31 years (1979–2010), until the plant's relocation in 2010. Additionally, water inrush incidents from the Ordovician limestone aquifer have been occurring intermittently at the Sangshuping Coal Mine and Magouqu Coal Mine. Moreover, many villages and enterprises also rely on OLW as their potable water source. The water withdrawal volumes from the Ordovician limestone aquifer by these enterprises and villages are all documented in official water accounting records, based on which the long-term variations of hydrodynamic field of OLW can be depicted.

The extraction history of OLW can be divided into three distinct phases based on the intensity of exploitation (drainage). The phase I (1971–1978) represents the natural state period with minimal anthropogenic disturbance to the OLW. The phase II (1979–2010) represents period of intensive exploitation of OLW in Hancheng power plant and several water inrush events of OLW in the whole mining area. The phase III (2011–2022) represents condition of limited exploitation in the southern zone and a big OLW inrush incident in Sangshuping coal mine in the northern zone, and also several water supply wells for domestic and industrial use.

## **3 Materials and methods**

### **3.1 Sampling collection**

A total of 104 OLW samples, which are mainly located within the southeastern belt of the mining area, were collected between 1971–2022, with some sampling points sampled multiple times (Fig. 2). There are 18 samples in phase I, 38 in phase II and 48 in phase III. The major ion measurements were conducted by the specialist agency in accordance with the standard testing methods. Major ions (Na<sup>+</sup>, K<sup>+</sup>, Ca<sup>2+</sup>, Mg<sup>2+</sup>, SO<sub>4</sub><sup>2-</sup>, Cl<sup>-</sup>) were tested by ion chromatography with the precision of 1%. The traditional titration with HCl were used to measure HCO<sub>3</sub><sup>-</sup> and CO<sub>3</sub><sup>2-</sup>. Total dissolved solids (TDS) were calculated by the concentration of major ions. The accuracy and reliability of the analysis were verified by calculating the charge balance error (CBE) of each sample. The results for the samples showed a CBE less than 5%, indicating the reliability of the analysis results.

### **3.2 Numerical simulation**

The numerical simulation method demonstrates significant advantages in modeling the evolution of groundwater flow fields. It can handle complex boundary conditions, aquifer heterogeneity and anisotropy (Chang et al. 2020; Karatzas 2017; Zhao et al. 2019), furthermore it can comprehensively account for precipitation recharge, riverine recharge, anthropogenic extraction and leakage recharge. Numerical simulation enables relatively accurate characterization of the OLW flow field under varying spatiotemporal extraction intensities. Crucially, precise delineation of the flow field serves as the fundamental basis for hydrogeochemical analysis. Without flow field simulation, hydrogeochemical interpretations risk being built upon erroneous hydrodynamic foundations, potentially leading to incorrect water chemistry conclusions.

Therefore, this study employs Visual MODFLOW software to simulate the hydrodynamic field under long-term Ordovician groundwater extraction, establishing a robust foundation for subsequent analysis of hydrochemical evolution.

### 3.3 Statistic analysis

The concentrations of major ions and total dissolved solids (TDS) in water serve as robust indicators of groundwater quality. Therefore, in order to figure out the temporal and spatial variation of the OLW quality, the average value of major-ion concentration and TDS in each phase of the whole mining area, as well as its northern and southern zones were calculated. Integrating concentration trends and water type variations provides enhanced insights into the Ordovician groundwater's chemical evolution. Further coupling ionic proportional relationships and mineral saturation indices (SI) allows for robust interpretation of groundwater formation mechanisms.

### 3.4 Hydrogeochemical modeling

#### 3.4.1 mixing modeling

Mixing process generates groundwater with unique chemistry through the mixing of two compositionally distinct water end-members. Consequently, given known chemical compositions of end-member waters prior to mixing and the resultant mixed groundwater, mixing simulations provide a robust means to determine exact mixing ratios (Hiscock, 2005). According to previous hydrogeological surveys (Lyu GL, 2003; Ma XD 2020), originally, the OLW was recharged mainly by surface water (rainfall and river) in its eastern outcrop area. While, after prolonged over-exploitation and dewatering, hydrodynamic field changed fundamentally (Fig. 5b and c), subjected to limited river discharge and rainfall availability, the OLW is then primarily recharged by regional lateral runoff from the northwestern buried area. Therefore, the shallow OLW in the southeastern mining area primarily receives mixed recharge from lateral deep OLW and surface water. Accordingly, the mixing ratio of deep OLW and surface water can be calculated according to Eq. (1).

$$C_{i,mix} = f_d \bullet C_{i,d} + f_s \bullet C_{i,s}$$

1

Where  $C_{i,mix}$  is the concentration of ion  $i$ ;  $mix, d$  and  $s$  indicate the conservative mixture and end-member deep OLW and surface waters, and  $f$  is the fraction of end-member waters.

#### 3.4.2 Inverse modeling

Hydrogeochemical modeling enables the simulation of aqueous geochemical processes along flow paths (De Windt et al., 2025; Rafiq et al., 2022), including dissolution/precipitation, redox reactions, mixing processes, and ion exchange et al. Inverse modeling solves for stoichiometric reaction networks and associated mass transfers between constrained initial and evolved water compositions. The precondition for establishing an inverse model is to determine the chemical reactions occurring in the groundwater system. Therefore, chemical component analysis in the initial and final solutions are conducted to determine the chemical reactions. The basic idea of the inverse modelling is as follows:

Initial water chemistry + reactants = final water chemistry + products (2)

The “reactants” and “products” in Eq. (2) are substances entering or leaving the solution phase during the reaction process. Generally, various combinations of the “reactants” and “products” can satisfy Eq. (2). It is necessary to determine the most likely “reactants” and “products” based on the lithology, geology and hydrogeological conditions of the study area.

### 3.4.3 Chlorine alkalinity indices

Schoeller (1965) proposed two chlorine alkalinity indices (CAI-I and CAI-II) for the study of cation exchange. They are defined as follows:

$$CAI - I = \frac{Cl^{-} - (Na^{+} + K^{+})}{Cl^{-}}$$

3

$$CAI - II = \frac{Cl^{-} - (Na^{+} + K^{+})}{HCO_3^{-} + SO_4^{2-} + CO_3^{2-} + NO_3^{-}}$$

4

Positive values of CAI-I and CAI-II indicate ion exchange, namely the  $Ca^{2+}$  and  $Mg^{2+}$  in the surrounding strata have taken the place of  $Na^{+}+K^{+}$  in the solution (as shown in Eq. 3), while negative values indicate reverse ion exchange (as shown in Eq. 4).

## 4 Results and discussion

### 4.1 Hydrodynamic field variation

Due to differences in the extraction intensity and water inrush intensity of OLW between the southern and northern zones during various phases, the hydrodynamic field exhibits distinct characteristics across different stages. This study focuses solely on the numerical simulation results of the hydrodynamic field without elaborating on the simulation process. Figure 5 clearly illustrates the characteristics of the OLW hydrodynamic field across the three phases.

In phase I (Fig. 5a), the OLW flows from SW, where the water table was around 385 m, to NE, where the water table is around 376. While after decades of extraction (1979–2010) by the Hancheng Power Plant in phase II, a remarkably extensive groundwater depression cone centered on the plant had formed in the Ordovician limestone aquifer, with the water level at the cone's center declining by approximately 30 meters. This has driven centripetal groundwater flow from the southern periphery toward the extraction center, with concomitant water table decline observed even in the northern zone due to inter-regional hydraulic connectivity (Fig. 5b). At the early stage of phase III (2011–2022), a depression cone centered Sangshuping Mine in the northern zone had developed and the water table declined about 10 m in the cone center (Fig. 5c). Concurrently, the extensive southern cone demonstrated partial hydraulic recovery while

maintaining its structural persistence. Post-2011, the cessation of large-scale OLW extraction and water-inrush in the whole region allowed groundwater level recovery in the depression cones, driven by combined regional lateral recharge and surface water replenishment.

Long-term extraction and drainage have fundamentally altered the flow field characteristics of the Ordovician limestone groundwater over the past 51 years. A comprehensive understanding of these flow field variations enables accurate hydrogeochemical analysis.

## 4.2 Hydrochemical characteristics

### 4.2.1 Chemical composition and hydrochemical facies

Table 1 shows the average value of major-ion concentration and TDS in each phase of the whole mining area, as well as its northern and southern zones were calculated, and also the increment between each phase.

Table 1  
Average mass concentration of major ion and TDS of OLW (unit: mg/l)

| Range             | Time      | Type      | Na <sup>+</sup> +K <sup>+</sup> | Ca <sup>2+</sup> | Mg <sup>2+</sup> | Cl <sup>-</sup> | SO <sub>4</sub> <sup>2-</sup> | HCO <sub>3</sub> <sup>-</sup> | TDS     |
|-------------------|-----------|-----------|---------------------------------|------------------|------------------|-----------------|-------------------------------|-------------------------------|---------|
| Whole mining area | I         | Average   | 122.99                          | 117.13           | 39.72            | 122.80          | 317.82                        | 281.51                        | 861.21  |
|                   | II        | Average   | 230.43                          | 191.43           | 61.41            | 251.52          | 597.40                        | 305.52                        | 1484.95 |
|                   | III       | Average   | 611.85                          | 242.32           | 73.63            | 654.83          | 893.03                        | 429.68                        | 2690.51 |
|                   | I to II   | Increment | 0.87                            | 0.63             | 0.55             | 1.05            | 0.88                          | 0.09                          | 0.76    |
|                   | II to III | Increment | 1.66                            | 0.27             | 0.20             | 1.60            | 0.49                          | 0.41                          | 0.81    |
| Southern Zone     | I         | Average   | 64.31                           | 103.82           | 36.27            | 59.00           | 285.25                        | 199.86                        | 648.58  |
|                   | II        | Average   | 302.5                           | 218.11           | 75.28            | 311.23          | 801.17                        | 284.89                        | 1850.74 |
|                   | III       | Average   | 462.51                          | 257.73           | 93.84            | 492.06          | 1018.14                       | 322.20                        | 2485.37 |
|                   | I to II   | Increment | 3.70                            | 1.10             | 1.08             | 4.28            | 1.81                          | 0.43                          | 1.85    |
|                   | II to III | Increment | 0.53                            | 0.18             | 0.25             | 0.58            | 0.27                          | 0.13                          | 0.34    |
| Northern Zone     | I         | Average   | 196.34                          | 130.20           | 43.97            | 183.63          | 349.44                        | 358.16                        | 1082.66 |
|                   | II        | Average   | 183.42                          | 174.03           | 52.36            | 212.57          | 464.51                        | 318.98                        | 1246.39 |
|                   | III       | Average   | 749.24                          | 228.16           | 55.05            | 804.58          | 777.92                        | 528.56                        | 2879.23 |
|                   | I to II   | Increment | -0.07                           | 0.34             | 0.19             | 0.16            | 0.33                          | -0.11                         | 0.14    |
|                   | II to III | Increment | 3.08                            | 0.31             | 0.05             | 2.78            | 0.67                          | 0.66                          | 1.34    |

Visualizing the data from Table 1 graphically would provide more intuitive insights into the evolutionary trends of ion concentrations and TDS from Phase I to Phase III (Fig. 6). The average concentrations of chemical compositions in OLW show an obvious increase trend from 1971 to 2022. The average TDS of OLW in the whole mining area increased from 861.21 mg/l to 1484.95 mg/l and 2690.51 mg/l across the three phases. While that in the southern zone is 648.58 mg/l, 1850.74 mg/l and 2485.37 mg/l; and 1082.66 mg/l, 1246.39 mg/l and 2879.23 mg/l in the northern zone (Fig. 6a).

The  $\text{Na}^+\text{+K}^+$ ,  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$ ,  $\text{Cl}^-$ ,  $\text{SO}_4^{2-}$  and  $\text{HCO}_3^-$  in the whole mining area (Fig. 6b), the northern zone (Fig. 6c) and the southern zone (Fig. 6d) are all increased from phase I to phase III, except the  $\text{HCO}_3^-$  in southern zone from phase II to phase III, among which the increase of  $\text{SO}_4^{2-}$ ,  $\text{Cl}^-$  and  $\text{Na}^+\text{+K}^+$  contributes most to the rise of TDS.

The increments of the ionic concentration and TDS between two adjacent phases can effectively indicate the degree of groundwater evolution, which were calculated (Table 1) and figured (Fig. 7). The major-ion concentration in phase I-II showed a larger increase than phase II-III in the southern zone, whereas the northern zone exhibited the reverse pattern. A robust correlation between the rising rates of ionic concentrations and groundwater extraction intensities can be established on account of that in phase II the power plant (in the southern zone) extracting heavily while in phase III the Sangshuping Mine (in the northern zone) dewatering heavily. The hydrodynamic field characteristics (Fig. 5b and c) provide compelling evidence supporting this conclusion.

## 4.2.2 Water type

Hydrochemical types can also reflect the evolution characteristics of groundwater. A Piper diagram is plotted to represent the relative major-ion concentration of OLW in the three phases (Fig. 8). The hydrochemical types exhibit distinct differences in different phases, which is dominated by  $\text{HCO}_3\text{-Ca}$  and  $\text{SO}_4\text{-HCO}_3\text{-Ca}$  in phase I, and convert into  $\text{SO}_4\text{-HCO}_3\text{-Ca-Na}$  in phase II, and show a more complex type, with  $\text{Na}^+$  being the dominant cation and  $\text{SO}_4^{2-}$ ,  $\text{Cl}^-$ ,  $\text{HCO}_3^-$  constituting the primary anions, in phase III. The observed evolution of hydrochemical types can be inferred to result predominantly from the elevated concentrations of  $\text{Na}^+$ ,  $\text{Cl}^-$ , and  $\text{SO}_4^{2-}$ , as evidenced by their significantly higher increasing rates compared to other major ions.

## 4.3 Mixing modelling

Mixing simulation was used to figure out the recharge ratio of surface water (rainfall and river) and regional lateral runoff (termed as deep OLW). Since  $\text{Cl}^-$  has a high solubility, it is neither easy to form precipitation with other ions nor be adsorbed by mineral or rock particles, and it is a more stable ion in groundwater (Shu et al. 2022), therefore, the mixing ratio of deep OLW and surface water can be calculated by  $\text{Cl}^-$  according to Eq. (1) and assuming that the mixing is mechanical (Xing et al. 2018). The average mass concentration of  $\text{Cl}^-$  in OLW samples in the southern and northern zones during each phase is taken as the  $\text{Cl}^-$  concentration in the mixed water, while the  $\text{Cl}^-$  concentrations in surface water and deep OLW are regarded as the end-member concentrations before mixing (Table 2), based on which the recharge ratio of surface water and

deep OLW was calculated (Table 3, Fig. 9). In view of the distinct hydrodynamic conditions in the three phases, recharge ratios of the OLW must be individually quantified.

Table 2  
Concentration of  $\text{Cl}^-$  in surface water, deep OLW and mixing water unit: mg/L

| Waters        | Phase I       |               | Phase II      |               | Phase III     |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|               | Southern zone | Northern zone | Southern zone | Northern zone | Southern zone | Northern zone |
| Surface water | 23.00         | 11.98         | 23.00         | 11.98         | 23.00         | 11.98         |
| Deep OLW      | 647.81        | 462.62        | 647.81        | 462.62        | 647.81        | 462.62        |
| Mixing water  | 59.00         | 183.63        | 417.51        | 219.07        | 479.20        | 301.48        |

Table 3  
Recharge ration of surface water and deeply buried OLW

|                          | Southern zone |          | Northern zone |          |
|--------------------------|---------------|----------|---------------|----------|
|                          | Surface water | Deep OLW | Surface water | Deep OLW |
| Phase I                  | 0.94          | 0.06     | 0.62          | 0.38     |
| Phase II                 | 0.37          | 0.63     | 0.55          | 0.45     |
| Phase III                | 0.27          | 0.73     | 0.36          | 0.64     |
| Increment from I to II   | -0.61         | 9.50     | -0.12         | 0.20     |
| Increment from II to III | -0.27         | 0.16     | -0.35         | 0.41     |

It is clearly evident that in the southern zone, the surface water accounted for 94% of the total recharge in phase I, while it decreased to 37% and 27% in phase II and phase III, respectively (Fig. 9a). Correspondingly, the recharge ration of deep OLW increased rapidly from 6% in phase I to 63% and 73% in phase II and III respectively. In regard to the northern zone (Fig. 9b), the variation trend of recharge ratio mirrors that of the southern zone, demonstrating a consistent pattern where surface water contribution progressively decreases and the deep OLW recharge proportion continuously increases. A strong correlation exists between periods of significant deep OLW recharge and phases of intensive shallow OLW over-exploitation. Specifically, in the southern zone intensive pumping during phase II led to the highest increase in deep OLW recharge ratio in this phase, while in the northern zone large dewatering volume in phase III resulted in the peak recharge ratio of deep OLW during this period.

## 4.4 Inverse modelling

The increase in the recharge proportion of deep OLW leads to the rise in TDS. Therefore, it is essential to clarify the hydrogeochemical processes occurring during the flow of OLW from shallow southeastern to deep northwestern zones, i.e., the spatial evolution patterns of OLW.

# 4.4.1 Water samples

Hydrochemical evolution from recharge to stagnant zones was investigated through hydrogeochemical modeling, utilizing temporally sequential samples from identical water inflow point to represent respective zone characteristics. The waters sampled at the No.7 water inflow point in Sangshuping mine in 1988 and 2022 are selected to analyze the main hydrogeochemical process along the SE-NW direction. Of which the 1988 sample represents recharge zone characteristics under minimal anthropogenic influence, while the 2022 sample reflects stagnant zone water composition after extensive deep OLW recharge. The main constituents of the two water samples were listed in Table 4.

Table 4

The main constituents of water samples from No.7 water inflow point in Sangshuping coal mine (unit: mmeq/l)

| Sample    | Time | pH  | Na <sup>+</sup> +K <sup>+</sup> | Ca <sup>2+</sup> | Mg <sup>2+</sup> | Cl <sup>-</sup> | SO <sub>4</sub> <sup>2-</sup> | HCO <sub>3</sub> <sup>-</sup> | Hydrochemical types                      |
|-----------|------|-----|---------------------------------|------------------|------------------|-----------------|-------------------------------|-------------------------------|------------------------------------------|
| ☐         | 1988 | /   | 3.92                            | 8.13             | 4.29             | 3.01            | 8.91                          | 4.63                          | SO <sub>4</sub> ·HCO <sub>3</sub> -Ca·Mg |
| ☐         | 2022 | 7.6 | 31.13                           | 15.05            | 7.09             | 28.21           | 20.02                         | 5.00                          | Cl·SO <sub>4</sub> -Na·Ca                |
| Increment |      |     | 27.21                           | 6.93             | 2.80             | 25.20           | 11.11                         | 0.37                          |                                          |

# 4.4.2 Chemical component analysis

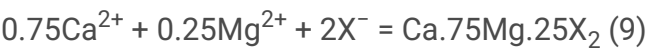
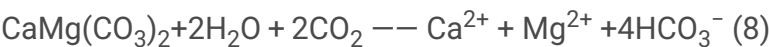
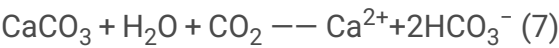
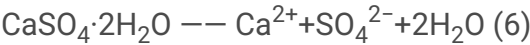
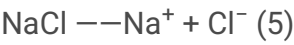
Generally, the Na<sup>+</sup> and Cl<sup>-</sup> in groundwater primarily derive from halite dissolution, exhibiting equimolar concentrations. While Table 4 shows the increment of equivalent concentration of Na<sup>+</sup>+K<sup>+</sup> is greater than that of Cl<sup>-</sup>, which may be caused by the cation exchange between Ca<sup>2+</sup> and Na<sup>+</sup> or K<sup>+</sup>. The existence of cation exchange may be confirmed by the Chloro-alkali index, proposed by Schoeller (Li et al. 2013; Marghade et al. 2012). The CAI (Eq. (3)~(4)), of which the CAI-I was - 0.3 and CAI-II was - 0.06, indicates that Ca<sup>2+</sup> and/or Mg<sup>2+</sup> have been removed from solution, and Na<sup>+</sup> and/or K<sup>+</sup> derived from surrounding rocks have taken their place. The strata of Ordovician in Hancheng mining area were primarily deposited in tidal flat environments, terrigenous detrital rocks were mixed in the carbonatite, which would contribute Na<sup>+</sup> and/or K<sup>+</sup> to the OLW. The equivalent concentration of Ca<sup>2+</sup>+Mg<sup>2+</sup> is less than SO<sub>4</sub><sup>2-</sup>+HCO<sub>3</sub><sup>-</sup>, from which it can be speculated that there are other sources of SO<sub>4</sub><sup>2-</sup> or Na and/or K had taken the place of Ca<sup>2+</sup> and/or Mg<sup>2+</sup>. Also, the increment of Ca<sup>2+</sup> (including Ca<sup>2+</sup>exchanged by Na<sup>+</sup> and/or K<sup>+</sup>) remains less than the sulfate increase, suggesting either additional SO<sub>4</sub><sup>2-</sup> sources or Ca<sup>2+</sup> removal via precipitation. Given the hydrologically confined setting of No.7 water inflow point, neither pyrite oxidation nor acid mine drainage contamination could occur. Therefore, Na and/or K took the place of Ca<sup>2+</sup> and/or Mg<sup>2+</sup> and the precipitation of calcite is more likely to happen. Further the SI of the main minerals in water samples ☐ and ☐ were calculated (Table 5). As can be seen from Table 5, calcite and dolomite transitioned from saturated conditions in 1988 to supersaturated states in 2022, whereas gypsum and halite remained undersaturated throughout both phases, albeit with higher SI in 2022 compared to 1988. Therefore, the difference between

the increment of  $\text{Ca}^{2+}$  and  $\text{SO}_4^{2-}$  is more likely caused by the precipitation of calcite and dissolution of gypsum.

Table 5  
SI of main mineral in OLW from No.7 water  
inflow point

| Mineral phase                                       | SI    |       |
|-----------------------------------------------------|-------|-------|
|                                                     | 1988  | 2022  |
| Calcite( $\text{CaCO}_3$ )                          | 0.09  | 0.23  |
| Dolomite( $\text{CaMg}(\text{CO}_3)_2$ )            | 0.03  | 0.25  |
| Gypsum( $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ ) | -0.88 | -0.52 |
| NaCl                                                | -6.64 | -4.83 |

Integrated analysis reveals the dominant hydrogeochemical processes affecting OLW from recharge to stagnant zone is that the dissolution of gypsum produced  $\text{Ca}^{2+}$ , then triggered by the common-ion effect of  $\text{Ca}^{2+}$ , the calcite precipitated, which would reduce the  $\text{Ca}^{2+}$  and  $\text{HCO}_3^-$ , promoting the dissolution of dolomite, which would increase the  $\text{Mg}^{2+}$  and  $\text{HCO}_3^-$ . Therefore, the main hydrogeochemical processes in the OLW from 1988 to 2022 can be identified as follows (Eq. (5)~(9)).



### 4.4.3 Result

Based on the above analysis, inverse modeling was conducted to quantify the mole transfer of main minerals in the chemical reactions and further to enhance the understanding of the OLW evolution in No.7 water inflow point.

The previous deduction was proved by the results of inverse modeling (Table 6), from which we can see that there are 5.81 mmol/l gypsum, 1.29 mmol/l dolomite and 25.32 mmol/l halite dissolved during 1988 to 2022, and 2.34 mmol/l calcite precipitated, and also 0.61g  $\text{CO}_2$  escaped. As for the cation exchange, there are 1.48 mmol/l  $\text{Ca}^{2+}$  removed from and 2.96 mmol/l  $\text{Na}^+$  entered into the OLW.

Table 6  
The mole transfers of main minerals in OLW from No.7 water inflow point (unit: mmol/l)

| Minerals       | Gypsum | Calcite | Dolomite | CO <sub>2</sub> /g | CaX <sub>2</sub> | NaX  | Halite |
|----------------|--------|---------|----------|--------------------|------------------|------|--------|
| Mole transfers | 5.81   | -2.34   | 1.29     | -0.61              | -1.48            | 2.96 | 25.32  |

The results of inverse modeling demonstrate firstly the increased recharge proportion of deep OLW, reflecting hydrodynamic field evolution, and secondly reveal the key hydrogeochemical processes governing water composition evolution along the recharge-to-stagnant zone pathway.

## 5 Conclusions

Long-term over-exploitation and dewatering have fundamentally altered the hydrodynamic regime in the Hancheng mining area. Pre-anthropogenic interference period, the OLW is recharged by the surface water infiltration through eastern outcrop area, while during intensive exploitation, the formation and progressive expansion of depression cones induced enhanced regional lateral inflow recharge, with artificial pumping/dewatering superseding natural discharge.

Surface water dominates the recharge source of OLW in phase I, which accounts for 94% for the southern zone, and 62% for the northern zone; while it declined to 37% and 55% in phase II, 27% and 36% in phase III. Accordingly, the recharge ratio of deep OLW is continuously increasing, which is 6% for the southern zone, and 38% for the northern zone in phase I; and it increased to 63% and 45% in phase II, 73% and 64% in phase III. The increasing proportion of deep OLW is the key reason for the deterioration of water quality in the Hancheng mining area.

The mass concentration of Na<sup>+</sup>+K<sup>+</sup>, Ca<sup>2+</sup>, Mg<sup>2+</sup>, Cl<sup>-</sup>, SO<sub>4</sub><sup>2-</sup>, HCO<sub>3</sub><sup>-</sup> and TDS are all increased from 1971 to 2022. As for the whole mining area the TDS increased progressively from 861.21 mg/l to 2690.51 mg/l during 1971 to 2022, marking a 212% rise, while its southern and northern zones exhibited the same trends and Na<sup>+</sup>+K<sup>+</sup>, Cl<sup>-</sup> and SO<sub>4</sub><sup>2-</sup> contribute most to the increase of TDS. Besides the largest increase of TDS in the southern zone occurred in phase II while that in the northern zone in phase III, which is well corresponding to the exploitation/dewatering intensity.

The main hydrogeochemical processes in OLW from recharge area to stagnant zone is the dissolution of gypsum, dolomite and halite, the precipitation of calcite and the cation exchange of Ca to Na.

## Declarations

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**Data availability** The data that support the findings of this study are available from the corresponding author, upon reasonable request.

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## Figures

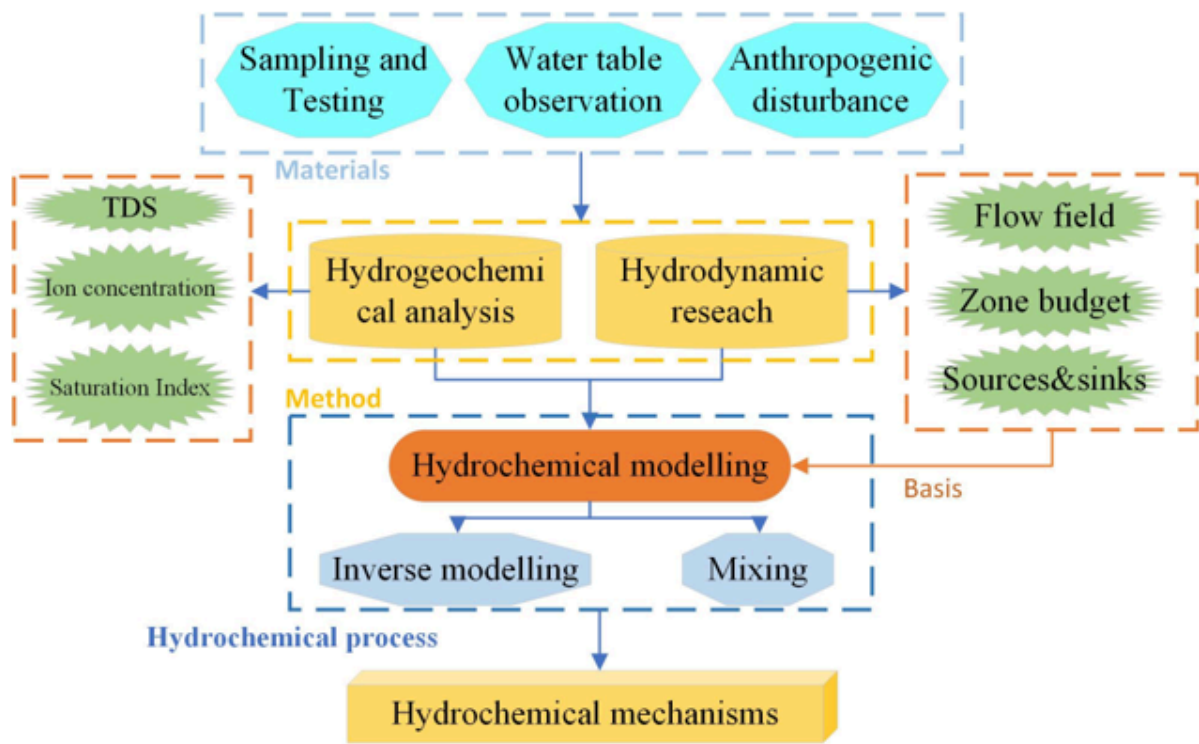
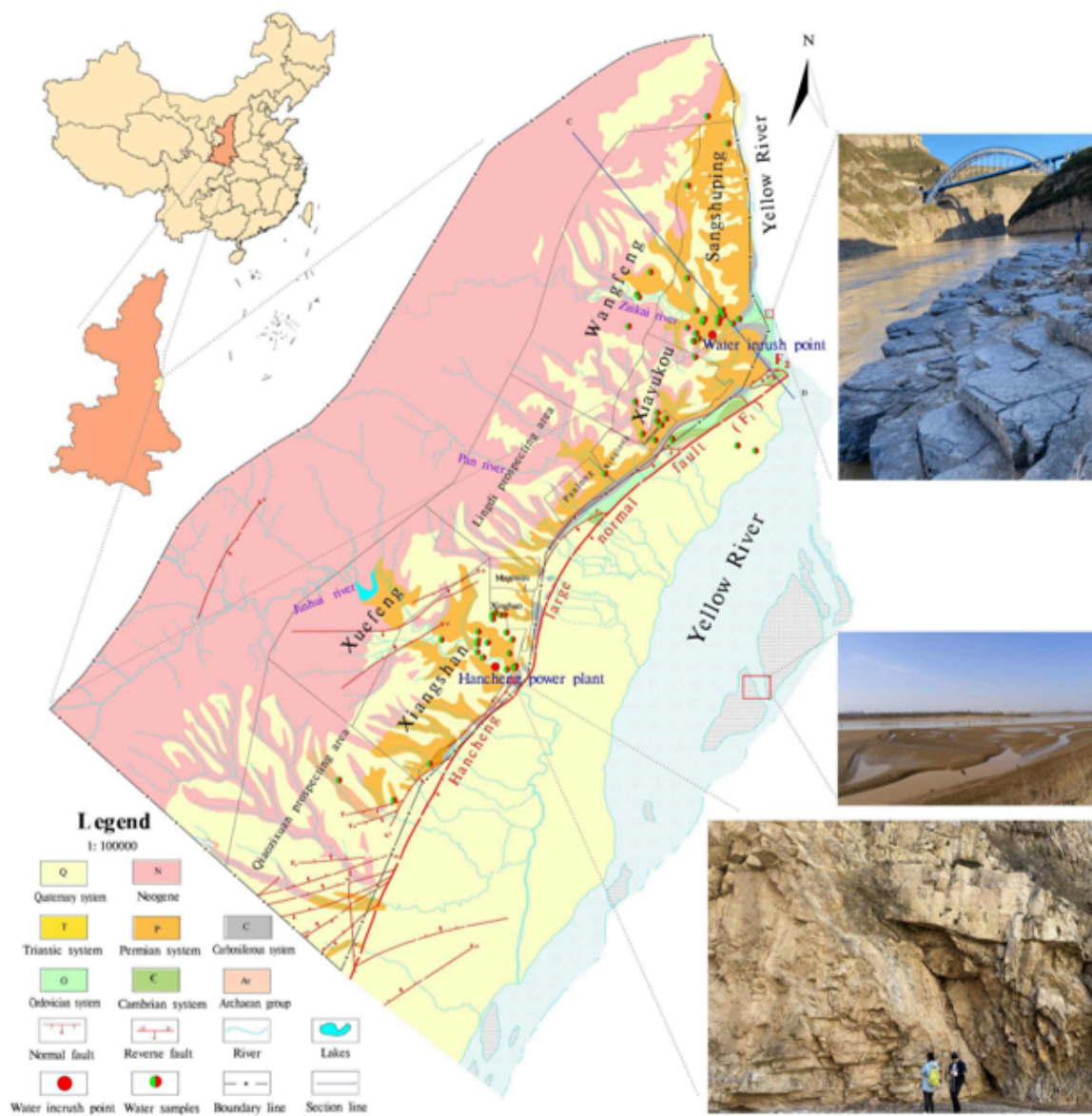


Figure 1

The flowchart of the research



**Figure 2**

Location of Hancheng mining area and the locations of groundwater samples

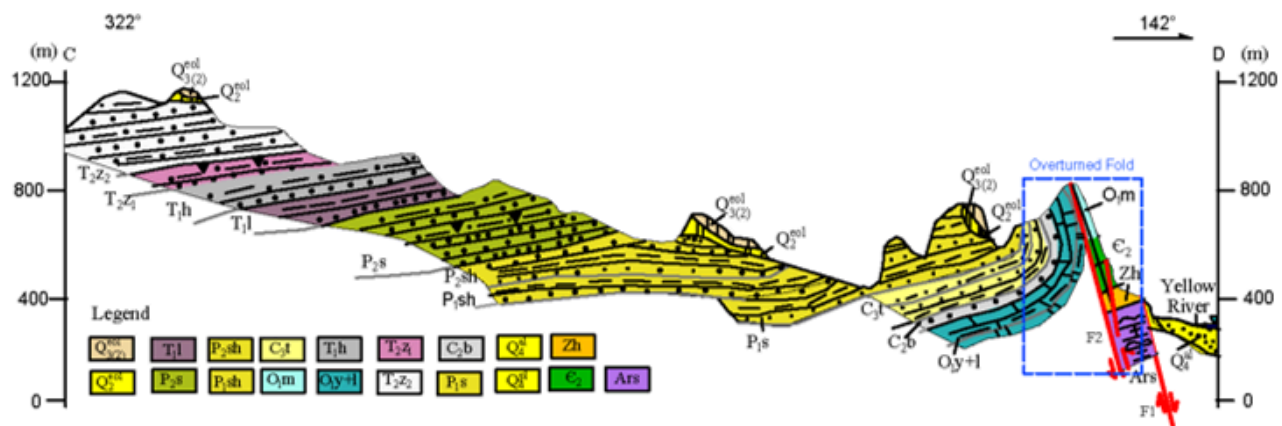


Figure 3

The geological profile of C-D in Fig. 1

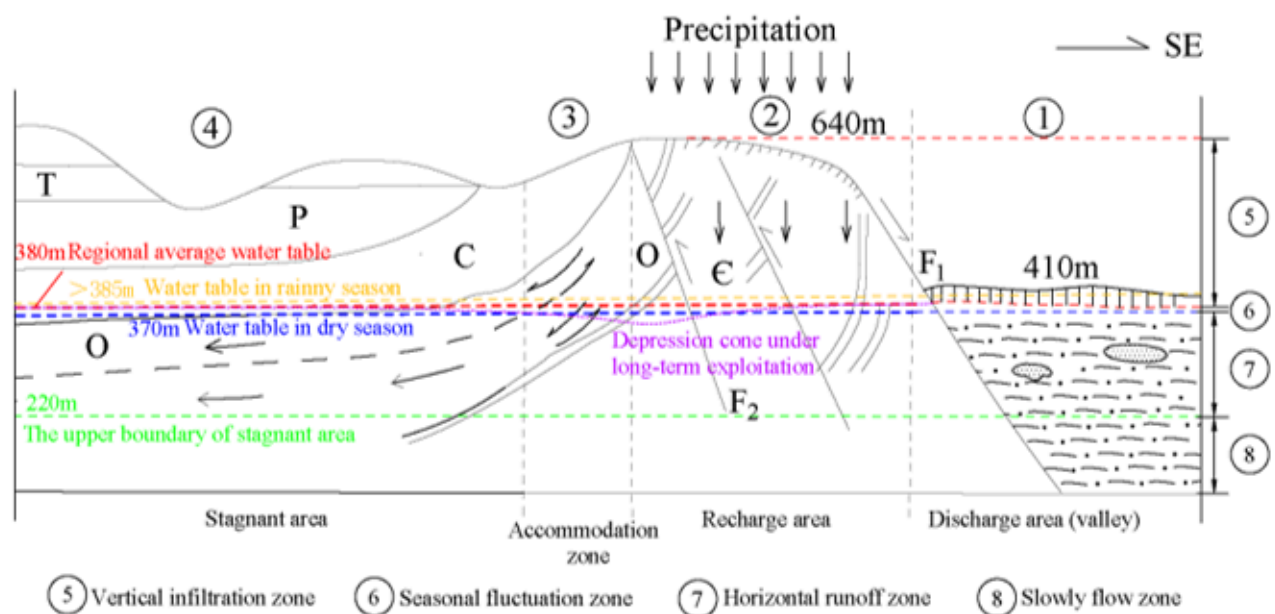
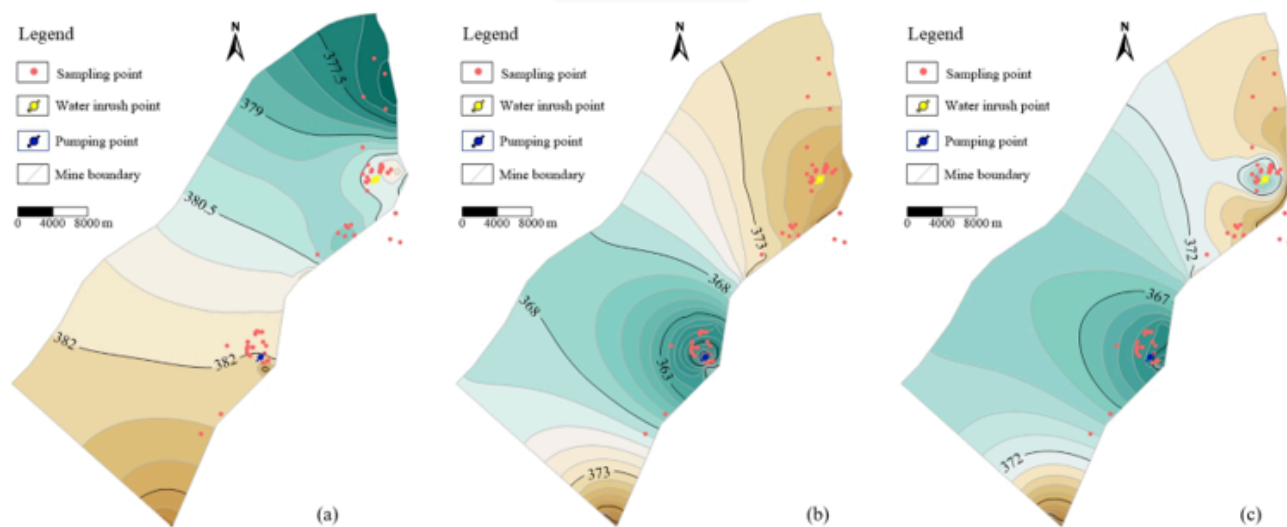


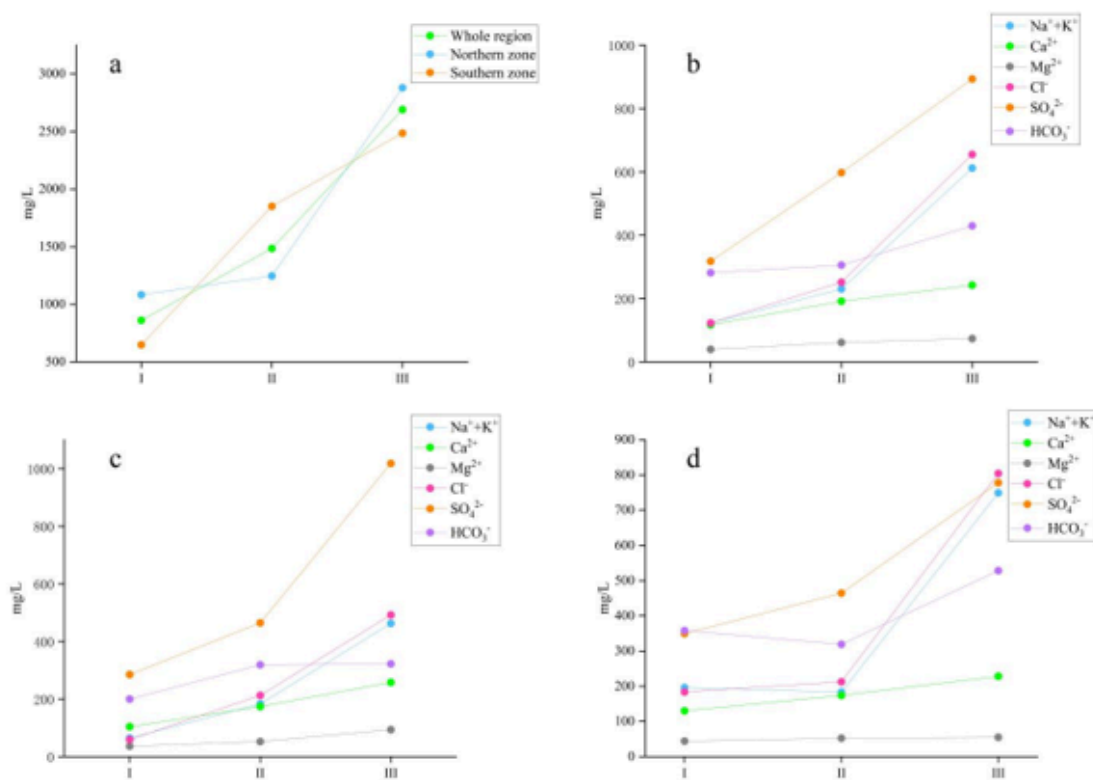
Figure 4

The conceptual model of OLW flow in NW-SE direction



**Figure 5**

Flow field of OLW in different phases. (a) phase I; (b) phase II; (c) phase III



**Figure 6**

The evolution of TDS and concentration of major ions from phase I to phase III. (a) The TDS of the whole mining area, northern and southern zones, (b) The whole mining area; (c) The southern zone; (d) The northern zone.

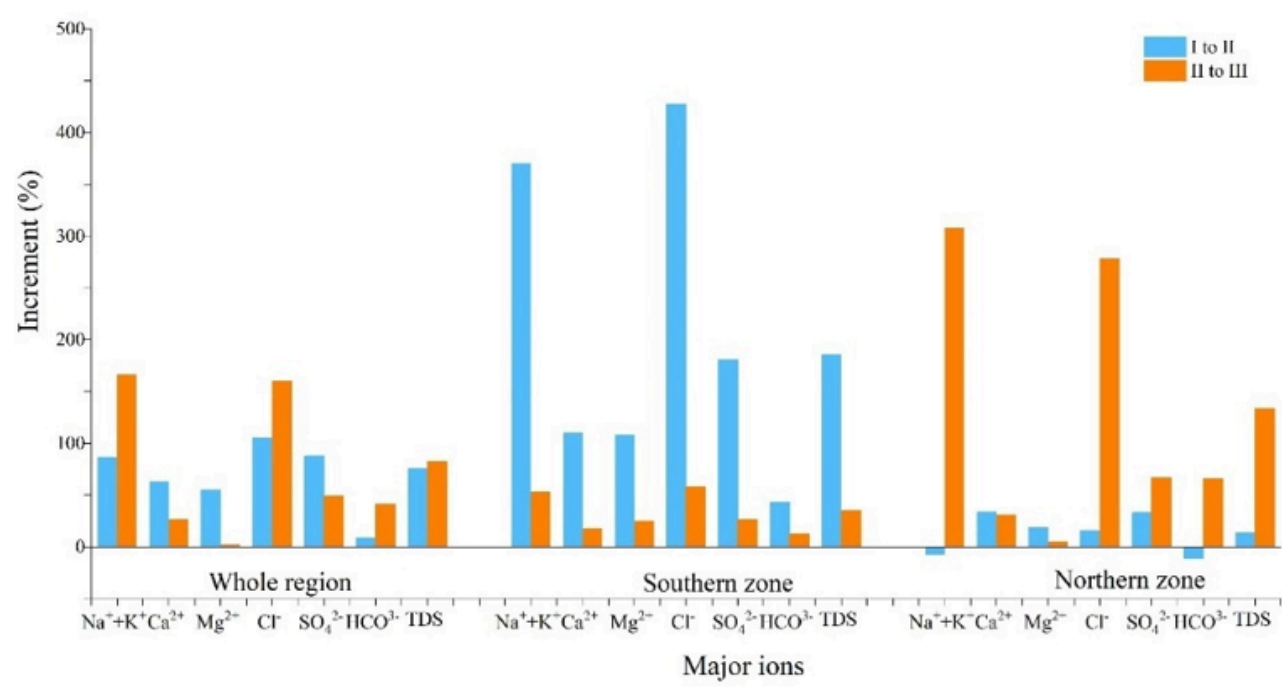


Figure 7

The increment of major-ion concentration of and TDS from phase I to phase III.

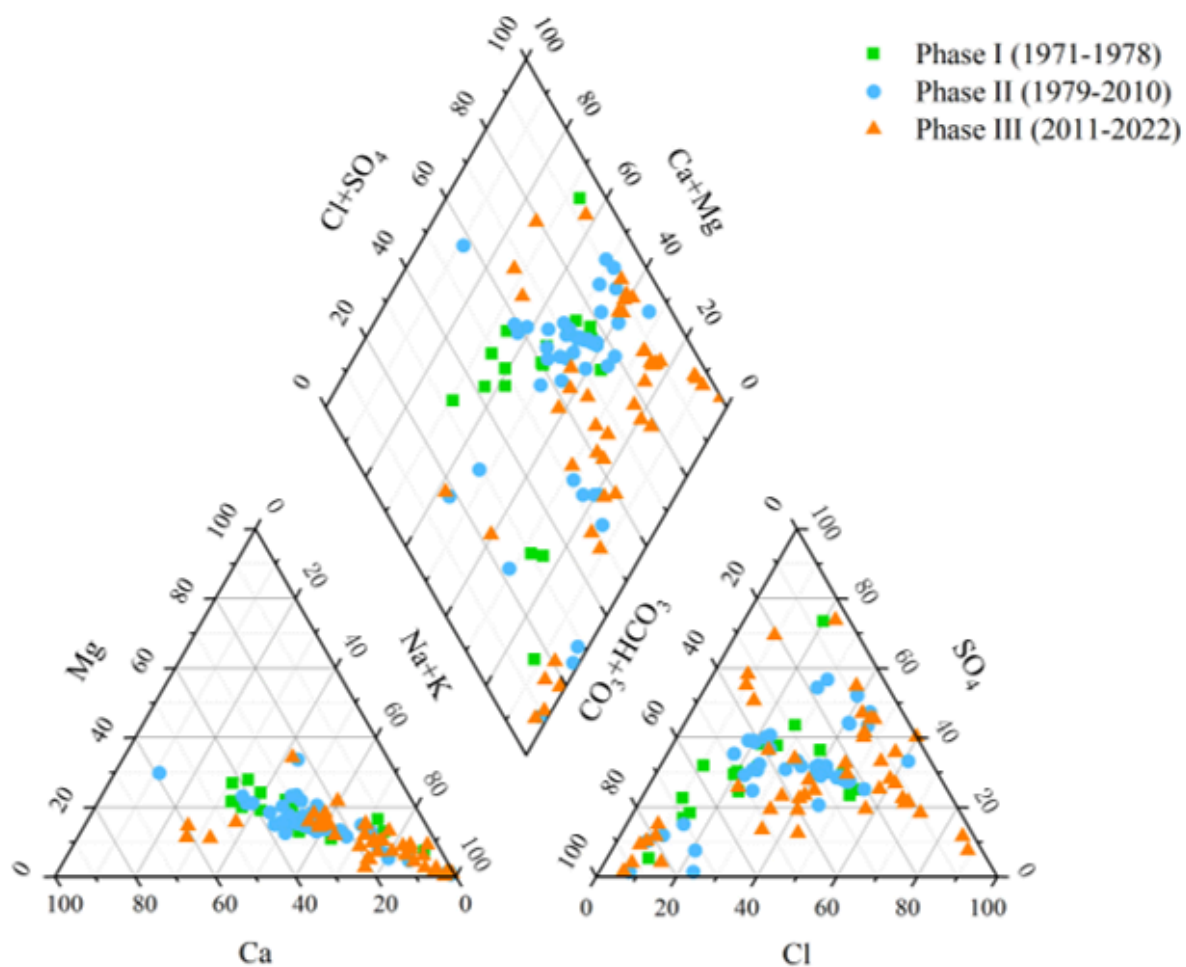


Figure 8

Piper diagram of OLW during 1971-2022

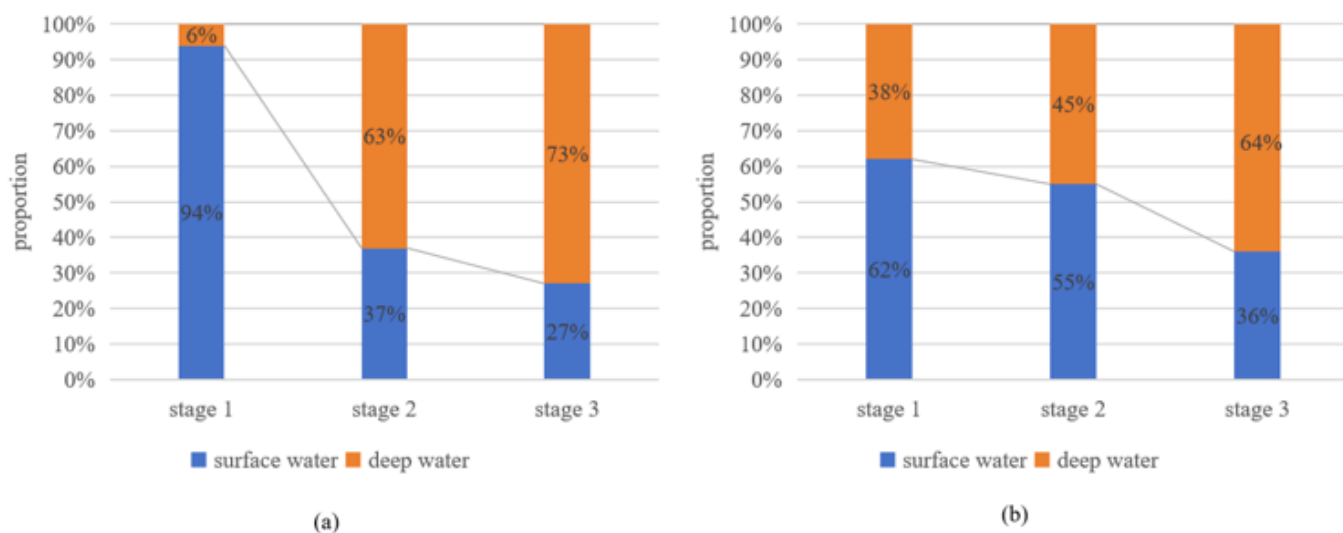


Figure 9

The variation of recharge ratio of surface water and deep water. (a) Southern zone; (b) Northern zone.